

Observation Task 1: Java Programming Fundamentals

2 Hours

Objective

To understand and implement the basic programming constructs of Java, including variables, data types, operators, conditional statements, loops, and methods.

Concepts Covered

- Java Program Structure
- Variables and Data Types
 - Operators
 - Input and Output
- Conditional Statements (if, if-else, switch)
- Looping Statements (for, while, do-while)
- Methods (User-defined Functions)

Task Description

Develop a "Student Utility Program" that performs the following operations using a menu-driven approach. Part A: Java Basics

- Display student information.
- Accept Name, Roll Number, and Marks.
- Calculate Total and Percentage.

•

Display the result.

Steps of the Program

Step 1: Import Scanner

```
import java.util.Scanner;
```

This is used to take input from the keyboard.

Step 2: Create the Student class

The class contains:

```
studentname  
rollno  
s1  
s2  
s3  
total  
percentage
```

These store the student's information and marks.

Step 3: Create studentdetails()

This method asks the user to enter:

```
Student name  
Roll number  
First subject marks  
Second subject marks  
Third subject marks
```

The entered values are stored in the corresponding variables.

Step 4: Create calc()

This method calculates the total:

```
total = s1 + s2 + s3;
```

Then calculates percentage:

```
percentage = total / 3.00;
```

Step 5: Create show()

This method displays:

```
Student name  
Roll number  
Subject 1 marks  
Subject 2 marks  
Subject 3 marks  
Total  
Percentage
```

Step 6: Create StudentProgram

The main() method creates a Student object:

```
Student st = new Student();
```

Step 7: Display the menu

The program repeatedly displays:

1. Enter student details
2. Calculate total and percentage
3. Show result
4. Exit

Step 8: Read the user's choice

```
choice = sp.nextInt();
```

Step 9: Execute the selected option

A switch statement handles the choice:

```
Choice  
Operation  
1  
Enter student details  
2  
Calculate total and percentage  
3  
Show result  
4  
Exit  
Other  
Invalid choice
```

Step 10: Repeat the menu

The do-while loop continues until:

```
choice == 4
```

Step 11: Exit

When the user selects 4, the program displays:

```
exited  
and stops.
```

START

1. Import Scanner.

2. Define the Student class.

3. Declare:
studentname
rollno
s1, s2, s3
total
percentage

4. Define studentdetails() method.

5. In studentdetails():
a. Read student name.
b. Read roll number.
c. Read first subject marks.
d. Read second subject marks.
e. Read third subject marks.

6. Define calc() method.

7. In calc():
a. Calculate:
 $total = s1 + s2 + s3$
b. Calculate:
 $percentage = total / 3.00$
c. Display total and percentage.

8. Define show() method.

9. In show():
Display student name.
Display roll number.
Display subject marks.
Display total.
Display percentage.

10. In main():
Create Scanner object.
Create Student object.

11. Display the menu.

12. Read user's choice.

13. If choice is 1:
Call studentdetails().

14. If choice is 2:
Call calc().

15. If choice is 3:
Call show().

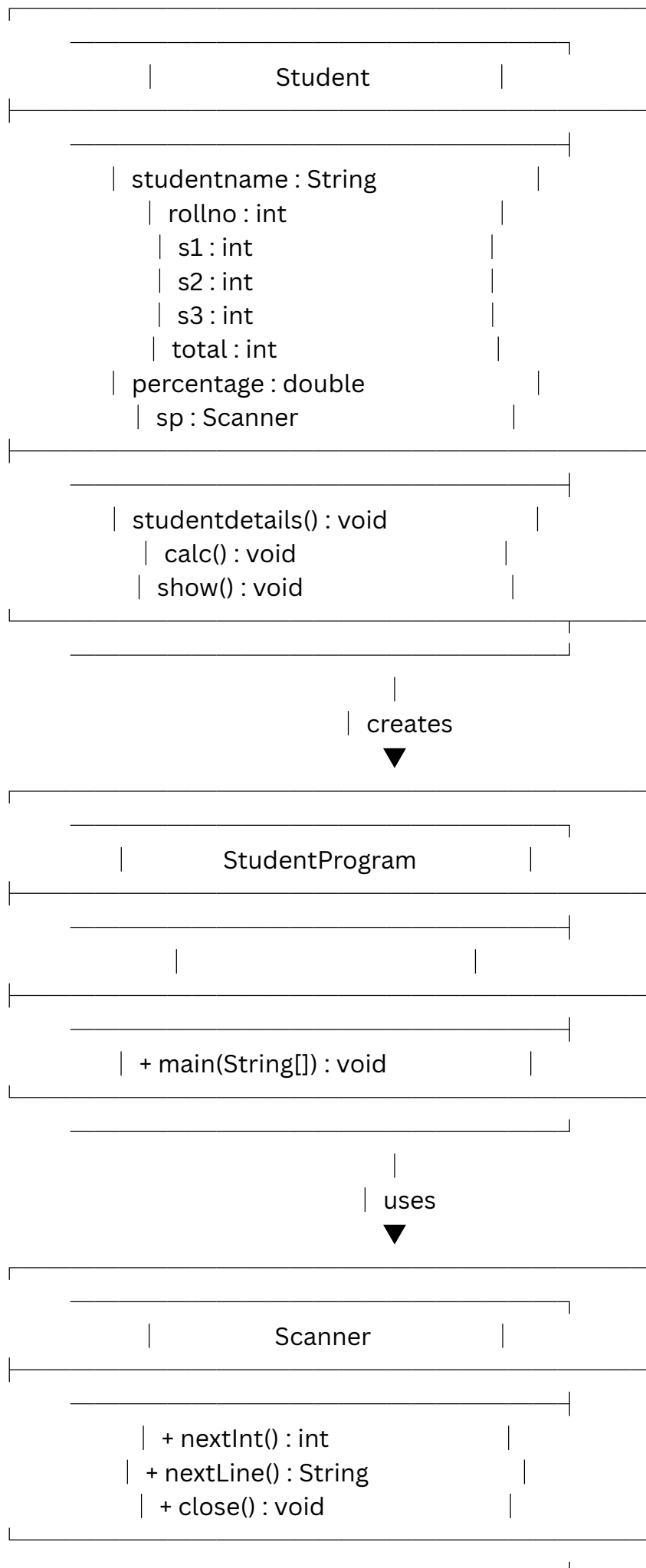
16. If choice is 4:
Display "exited".

17. Otherwise:
Display "invalid choice".

18. Repeat steps 11–17 until choice = 4.

19. Close Scanner.

20. STOP



```

import java.util.Scanner;

class Student{
    String studentname;
    int rollno;
    int s1,s2,s3;
    int total;
    double percentage;
    Scanner sp = new Scanner(System.in);
    void studentdetails(){
        System.out.println("enter the name of student");
        studentname = sp.nextLine();
        System.out.println("enter th roll number");
        rollno = sp.nextInt();
        System.out.println("enter the firstsubject marks");
        s1 = sp.nextInt();
        System.out.println("enter the secondsubject
            marks");
        s2 = sp.nextInt();
        System.out.println("enter the thirdsubject marks");
        s3 = sp.nextInt();
        sp.nextLine();
    }

    // total and percentage
    void calc(){
        total = s1+s2+s3;
        System.out.println("total is " + total);
        percentage = total/3.00;
        System.out.println("percentage is "+ percentage);
        System.out.println("the caluculation is completed");
    }

    void show(){
        System.out.println("----student details ----");
        System.out.println(" name of student is " +
            studentname);
        System.out.println(" roll number is " + rollno);
        System.out.println(" s1 marks is " + s1);
        System.out.println(" s2 marks is " + s2);
        System.out.println(" s3 marks is " + s3);
        System.out.println(" total marks is " + total);
        System.out.println(" total percentage is " + percentage);
    }
}

public class StudentProgram {
    public static void main(String[] args) {

        Scanner sp = new Scanner(System.in);
        Student st = new Student();
        int choice;
        do{
            System.out.println("--- student utility program---");
            System.out.println("1.enter student details ");
            System.out.println("2.calc total and percentage");
            System.out.println("3.show result");
            System.out.println("4.exit");
            System.out.println("enter your choice");
            choice = sp.nextInt();

            switch(choice){

                case 1:
                    st.studentdetails();
                    break;
                case 2:
                    st.calc();
                    break;
                case 3:
                    st.show();
                    break;
                case 4:
                    System.out.println("exited ");
                    break;
                default :
                    System.out.println("invalid choice");
            }
        }

        while (choice != 4) ;

        sp.close();
    }
}

```

javac StudentProgram.java
java StudentProgram

```
--- student utiility program----
    1.enter student details
    2.calc total and percentage
    3.show result
    4.exit
    enter your choice
        1
    enter the name of student
        Hemanth
    enter th roll number
        244
    enter the firstsubject marks
        85
enter the secondsubject marks
        90
    enter the thirdsubject marks
        80
--- student utiility program----
    1.enter student details
    2.calc total and percentage
    3.show result
    4.exit
    enter your choice
        2
        total is 255
        percentage is 85.0
    the caluculation is completed
--- student utiility program----
    1.enter student details
    2.calc total and percentage
    3.show result
    4.exit
    enter your choice
        3
    ----student details ----
    name of student is Hemanth
    roll number is 244
    s1 marks is 85
    s2 marks is 90
    s3 marks is 80
    total marks is 255
    total percentage is 85.0
--- student utiility program----
    1.enter student details
    2.calc total and percentage
    3.show result
    4.exit
    enter your choice
        4
    exited
```

Part B: Conditional Statements

- Check whether a number is Even or Odd.
 - Find the Largest of Three Numbers.
- Display Grade based on percentage using if-else.
 - Display Day of the Week using switch.

Program 1 — Even or Odd

Steps

Import the Scanner class.

Create a Scanner object.

Declare an integer n.

Ask the user to enter a number.

Read the number.

Check $n \% 2 == 0$.

If true, display even.

Otherwise, display odd.

Stop.

Program 2 — Largest of Three Numbers

Steps

Import Scanner.

Create a Scanner object.

Declare a, b, and c.

Read the three numbers.

Compare a with b and c.

If a is greater than or equal to both, print a.

Otherwise compare b with a and c.

If b is greater than or equal to both, print b.

Otherwise print c.

Stop.

Program 3 — Grade Based on Percentage

Steps

Import Scanner.

Create a Scanner object.

Declare a double variable called percentage.

Ask the user to enter the percentage.

Read the percentage.

If percentage is 90 or above, print GRADE - A.

Otherwise, if it is 80 or above, print GRADE - B.

Otherwise, if it is 70 or above, print GRADE - C.

Otherwise, if it is 60 or above, print GRADE - D.

Otherwise, print FAIL.

Stop.

Program 4 — Day of the Week

Your fourth program uses a switch statement.

Steps

Import Scanner.

Create a Scanner object.

Declare integer variable day.

Ask the user to enter a day number.

Read the day number.

Use a switch statement.

If day is 1, print Sunday.

If day is 2, print Monday.

If day is 3, print Tuesday.

If day is 4, print Wednesday.

If day is 5, print Thursday.

If day is 6, print Friday.

Otherwise print invalid day.

Stop.

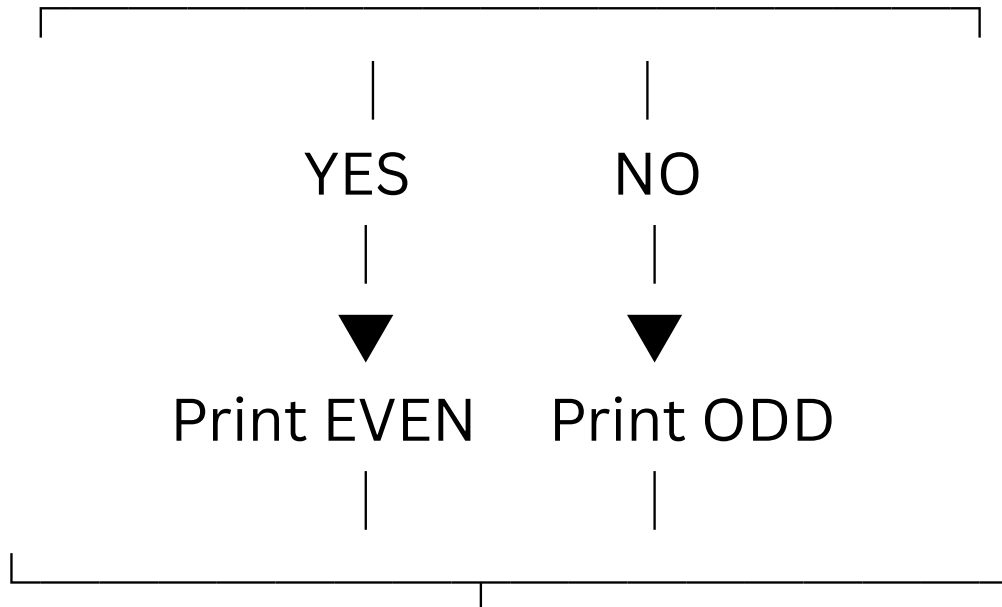
START



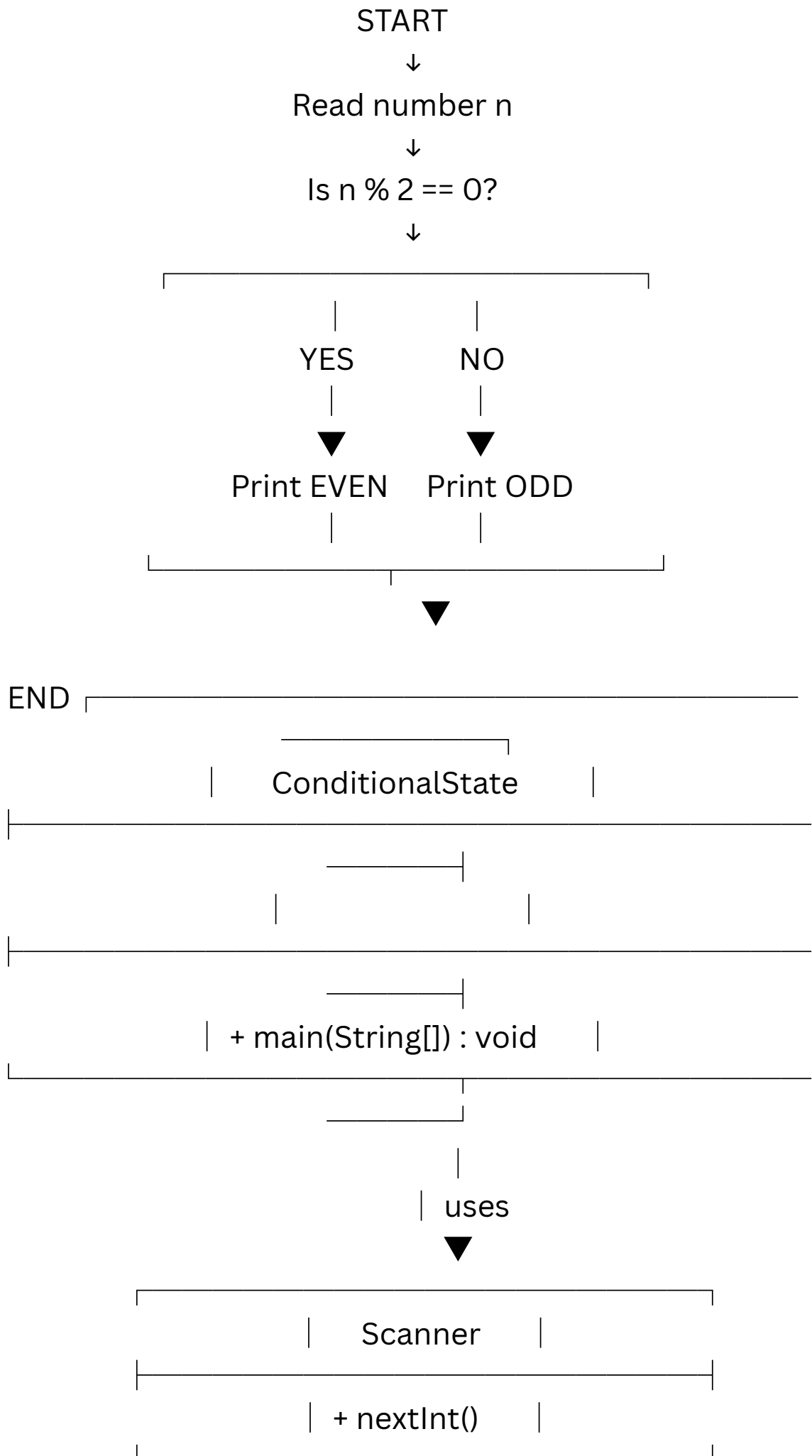
Read number n



Is $n \% 2 == 0$?



END



```
import java.util.Scanner;

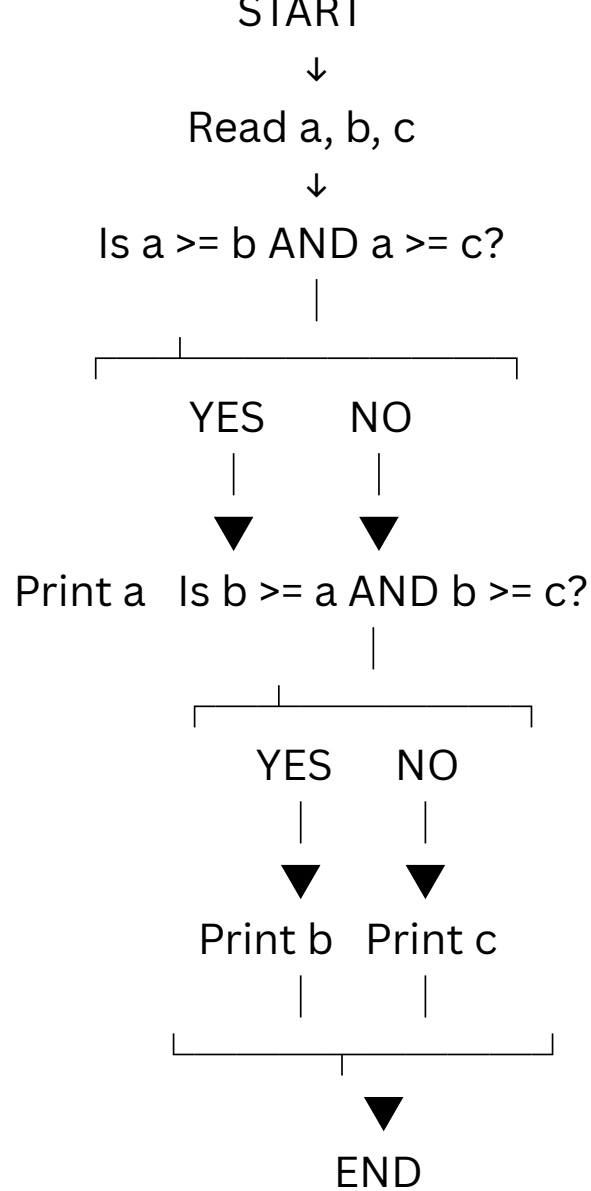
public class ConditionalState { // first program checks
    given number is even or odd
    public static void main(String[] args) {
        Scanner eo = new Scanner(System.in);
        int n;
        System.out.println("enter the number");
        n = eo.nextInt();
        if(n %2 == 0){
            System.out.println("the number is even");
        }
        else {
            System.out.println("the numder is odd");
        }
    }
}
```

```
javac ConditionalState.java
java ConditionalState
```

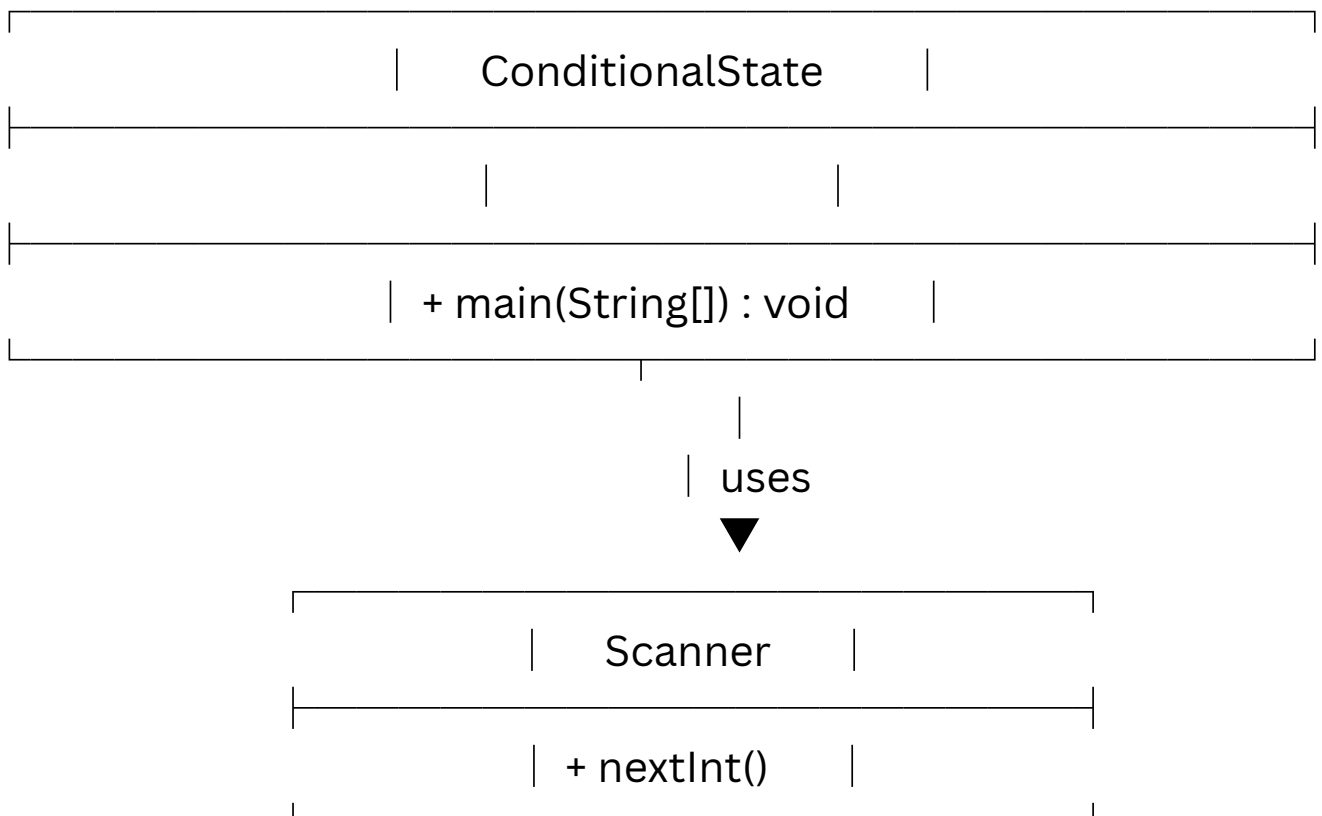
enter the number

7

the numder is odd



UML DIAGRAM



```

import java.util.Scanner;

public class ConditionalState {

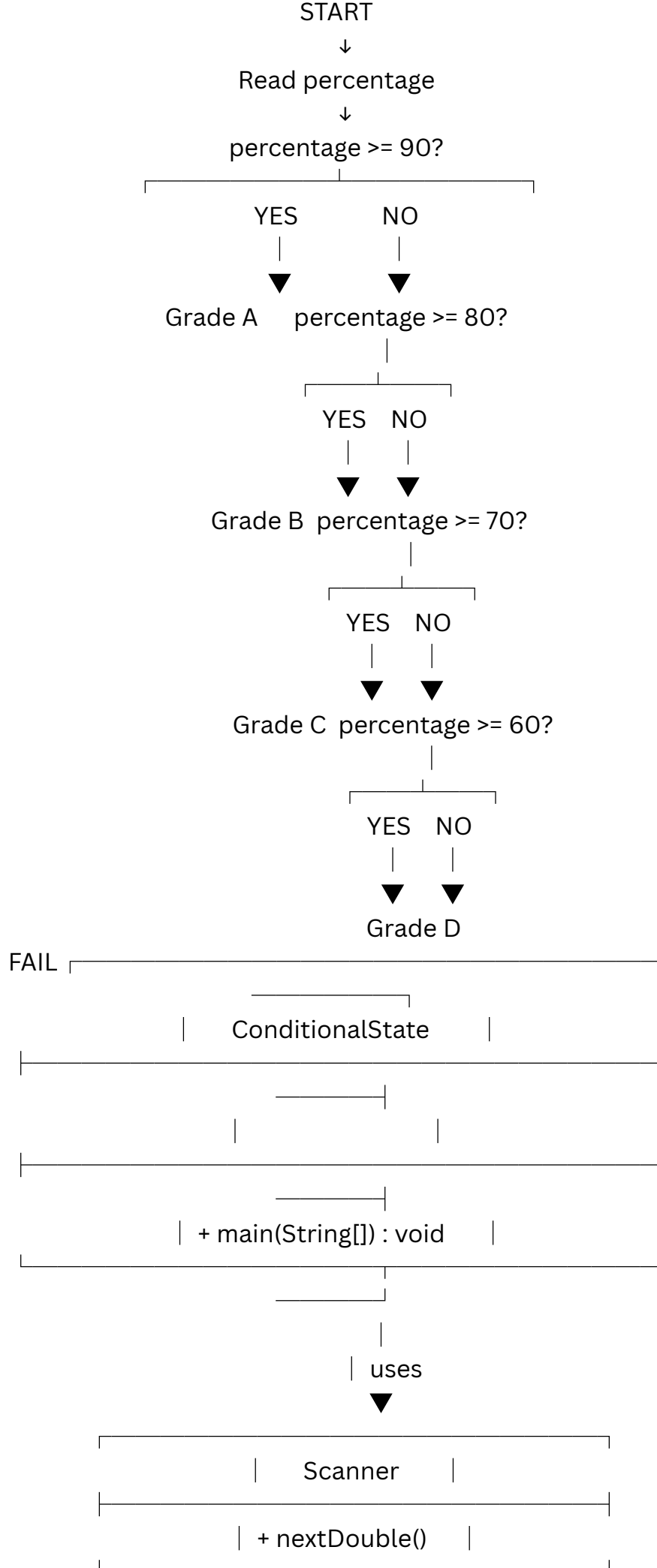
    public static void main(String[] args) {
        Scanner max = new Scanner(System.in);
        int a,b,c;
        System.out.println("enter the values of a,b,c");
        a = max.nextInt();
        b = max.nextInt();
        c = max.nextInt();
        if (a >= b && a >= c)
            {
                System.out.println("the largest number is " +a);
            }
        else if(b >= a && b >= c){
            System.out.println(" the largest number is " + b);
        }
        else{
            System.out.println("the largest number is " + c);
        }
    }
}

```

enter the values of a,b,c

10 25 15

the largest number is 25



```
import java.util.Scanner;

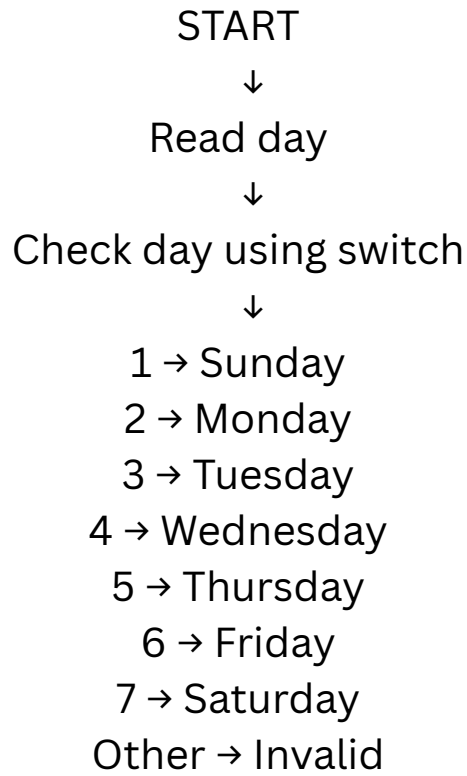
public class ConditionalState {
    public static void main(String[] args) {
        Scanner p = new Scanner(System.in);
        double percentage;
        System.out.println("enter the percentage");
        percentage = p.nextDouble();
        if (percentage >= 90)
            {System.out.println("GRADE - A");

            }else if (percentage >= 80)
                System.out.println("GRADE - B");
            else if (percentage >= 70)
                System.out.println("GRADE - C");
            else if (percentage >= 60)
                System.out.println("GRADE - D");
            else {
                System.out.println(" FAIL");
            }
        }
    }
}
```

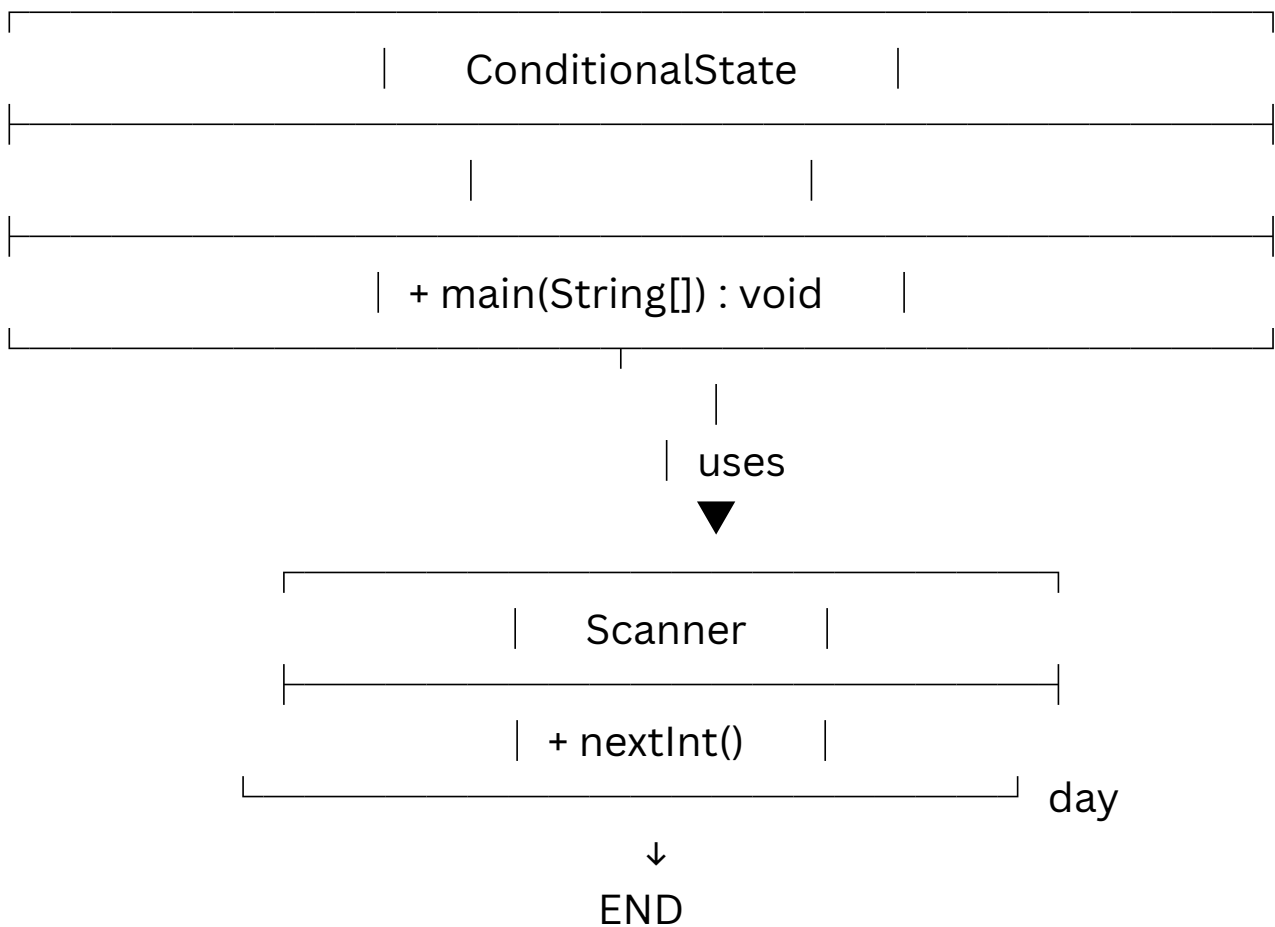
enter the percentage

85

GRADE - B



uml diagrams



```
import java.util.Scanner;
```

```
public class ConditionalState {  
    public static void main(String[] args) {
```

```
        Scanner d = new Scanner(System.in);
```

```
        int day;  
        System.out.println("enter the day");
```

```
        day = d.nextInt();
```

```
        switch (day) {  
            case 1:  
                System.err.println("sunday");  
                break;  
            case 2:  
                System.out.println("monday");  
                break;  
            case 3:  
                System.out.println("tuesday");  
                break;  
            case 4:  
                System.out.println("wednesday");  
                break;  
            case 5:  
                System.out.println("thrusday");  
                break;  
            case 6:  
                System.out.println("friday");  
                break;  
            default:  
                System.out.println("invalid day");  
                break;
```

```
        }
```

```
    }
```

```
}
```

enter the day

3

tuesday

Part C: Looping Statements

Write programs to:

- Print multiplication tables.
- Display numbers from 1 to N.
- Calculate the sum of first N natural numbers.
- Generate the Fibonacci series.

Steps
 Import Scanner.
 Create a Scanner object.
 Read a number n.
 Start a for loop from 1 to 10.
 Multiply n by the loop variable i.
 Display the result.
 Repeat until i = 10.
 Stop

ALGORITHM

START



Read n



Set i = 1



Is i <= 10?



YES



Calculate $n \times i$



Display result



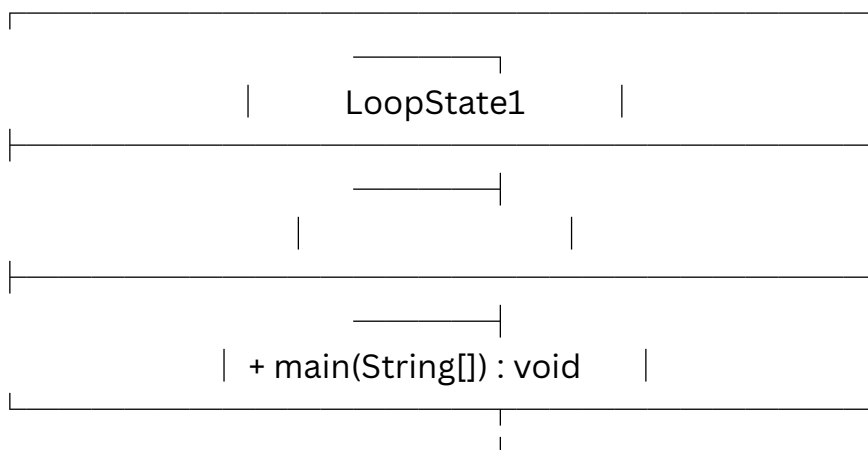
i = i + 1



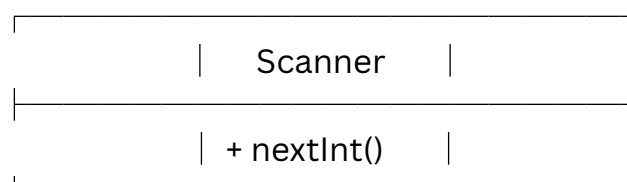
Check i <= 10



NO



uses



ENDprogram.

```
import java.util.Scanner;
```

looping statements of 1st program

```
public class LoopState {
```

```
    public static void main(String[] args) {
```

```
        Scanner m = new Scanner(System.in);
```

```
        int n;
```

```
        System.out.println("enter the number");
```

```
        n = m.nextInt();
```

```
        System.out.println("the multiplication table of given  
        number");
```

```
        for(int i = 1; i <= 10; i++){
```

```
            System.out.println(n + "x" + i + " = " + (n*i));
```

```
        }
```

```
    }
```

```
}
```

Enter the number

5

The multiplication table of given number

5 x 1 = 5

5 x 2 = 10

5 x 3 = 15

5 x 4 = 20

5 x 5 = 25

5 x 6 = 30

5 x 7 = 35

5 x 8 = 40

5 x 9 = 45

5 x 10 = 50

Steps
 Import Scanner.
 Create Scanner object.
 Read n.
 Start for loop with i = 1.
 Check whether i <= n.
 Display i.
 Increment i.
 Repeat until i becomes greater than

Algorithm

START



Read n



Set i = 1



Is i <= n?



YES



Display i



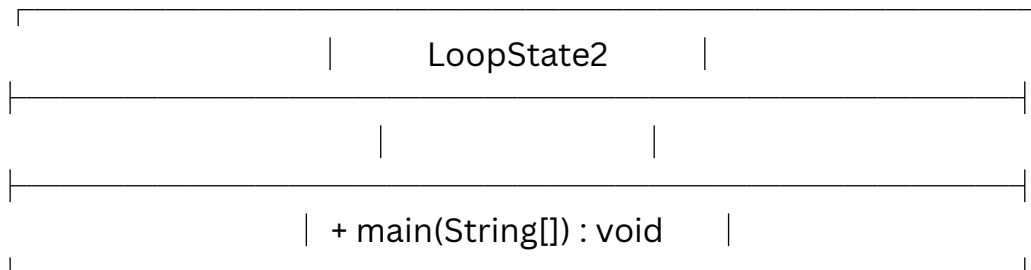
i = i + 1



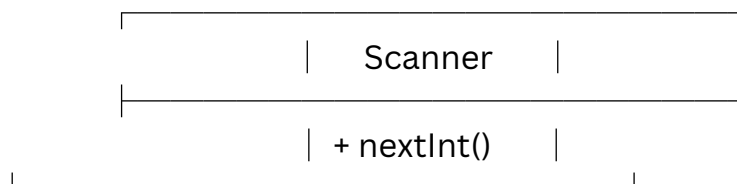
└──────────────────▶ Check condition



NO



| uses



ENDn.

Stop.

```
import java.util.Scanner;

public class LoopState {

    public static void main(String[] args) {
        Scanner N = new Scanner(System.in);
        int n;
        System.out.println(" enter the number");
        n = N.nextInt();
        System.out.println("Numbers from 1 to n" + (n));
        for (int i =1; i <= n; i++){
            System.out.println("");
        }
    }
}
```

Enter the number

5

Numbers from 1 to 5

1

2

3

4

5

Steps

Import Scanner.

Create Scanner object.

Read n.

Initialize sum = 0.

Start the loop from i = 1.

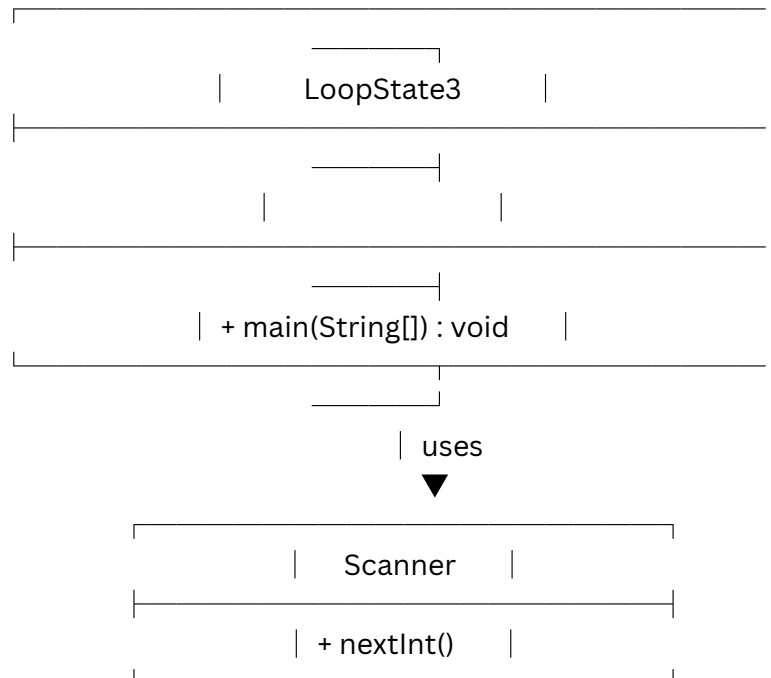
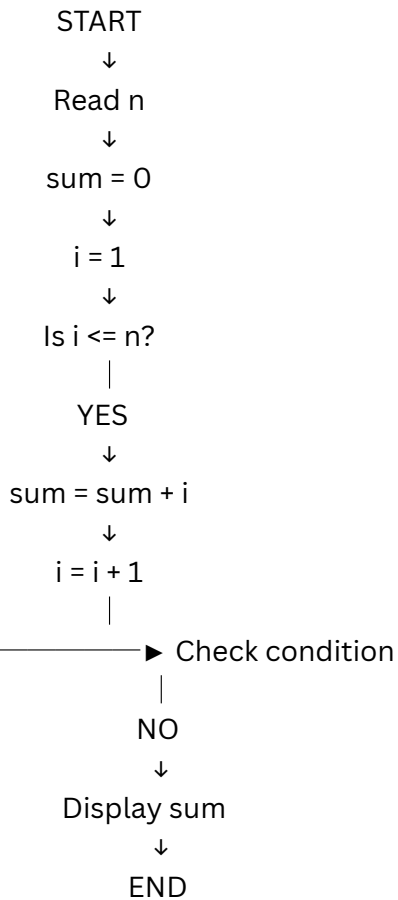
Add i to sum.

Increment i.

Continue until i = n.

Display the final sum.

Stop.




```
import java.util.Scanner;

public class LoopState {
    public static void main(String[] args) {
        Scanner sn = new Scanner(System.in);
        int n;
        System.out.println("enter the number");
        n = sn.nextInt();
        int sum = 0;
        for (int i = 1; i <= n; i++){
            sum = sum + i;
        }
        System.out.println("sum = " + sum);
    }
}
```

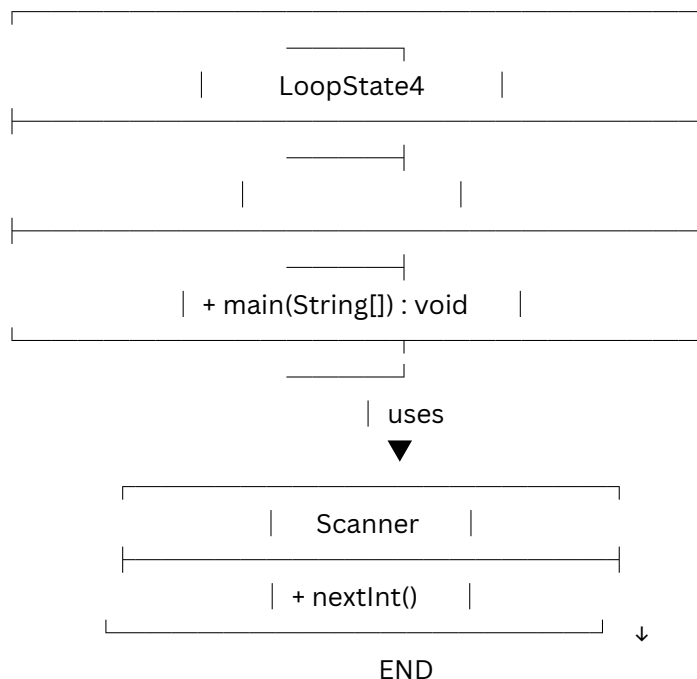
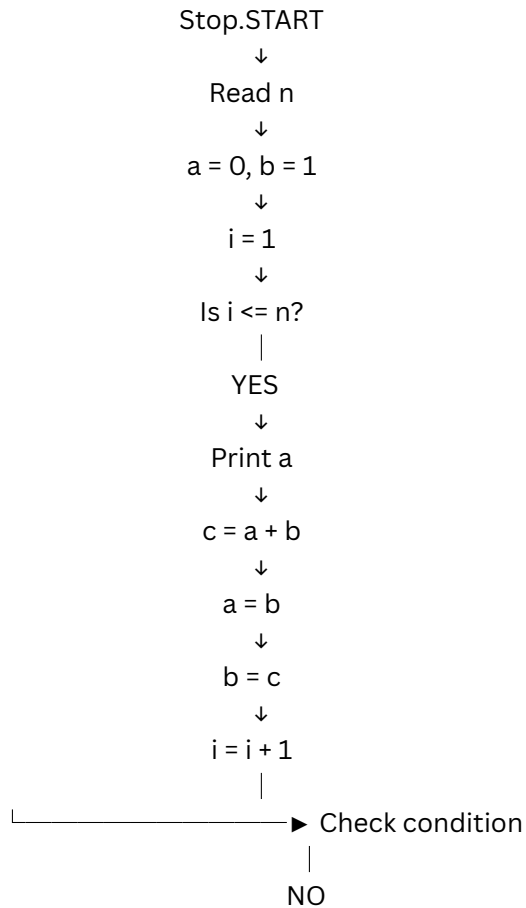
Enter the number

10

Sum = 55

Steps
 Import Scanner.
 Create Scanner object.
 Read the number of terms n.

Initialize:
 a = 0
 b = 1
 Start the for loop.
 Print a.
 Calculate:
 c = a + b
 Assign:
 a = b
 b = c
 Repeat until n terms are printed.



```
import java.util.Scanner;

public class LoopState {
    public static void main(String[] args) {
        Scanner f = new Scanner(System.in);
        int n;
        System.out.println("enter n");
        n = f.nextInt();
        int a = 0 ;
        int b= 1;
        System.out.println("the fibonacci series");
        for( int i= 1; i<= n;i++){
            System.out.print(a+"");

            int c = a+b;
            a = b;
            b = c;
        }
    }
}
```

Enter n

10

The Fibonacci series:

0 1 1 2 3 5 8 13 21 34

Part D: Methods : Every method should be parameterized and should return the result:

Create user-defined methods for:

- Finding the factorial of a number.
- Checking whether a number is prime.
- Finding the maximum of two numbers.
 - Calculating the area of a circle.

Steps

Start the program.

Import Scanner.

Create a method factorial(int n).

Initialize fact = 1.

Use a for loop from 1 to n.

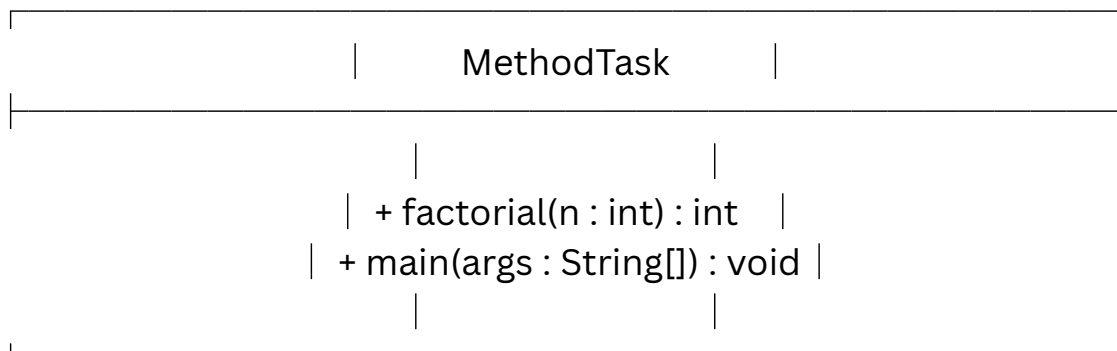
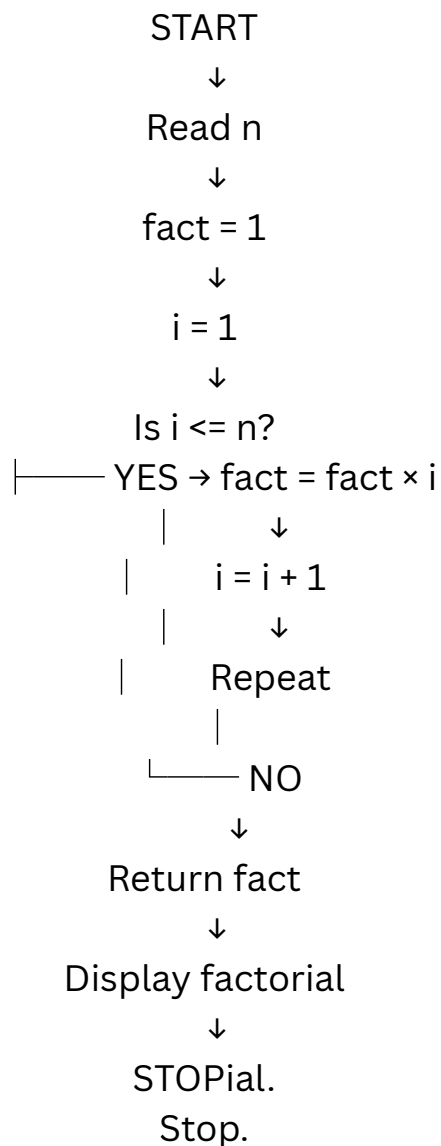
Multiply each number with fact.

Return fact.

In main(), read the number from the user.

Call factorial(n).

Display the factor



```
import java.util.Scanner;
// task on methods of 1st problem

public class MethodTask {
    static int factorial(int n){
        int fact =1 ;
        for(int i =1;i<= n;i++){
            fact = fact *i;
        }
        return fact;
    }
    public static void main(String[] args) {
        Scanner fa = new Scanner(System.in);
        int n , result;
        System.out.println("enter the number");
        n = fa.nextInt();
        result = factorial(n);
        System.out.println("factorial = " + result);

    }
}
```

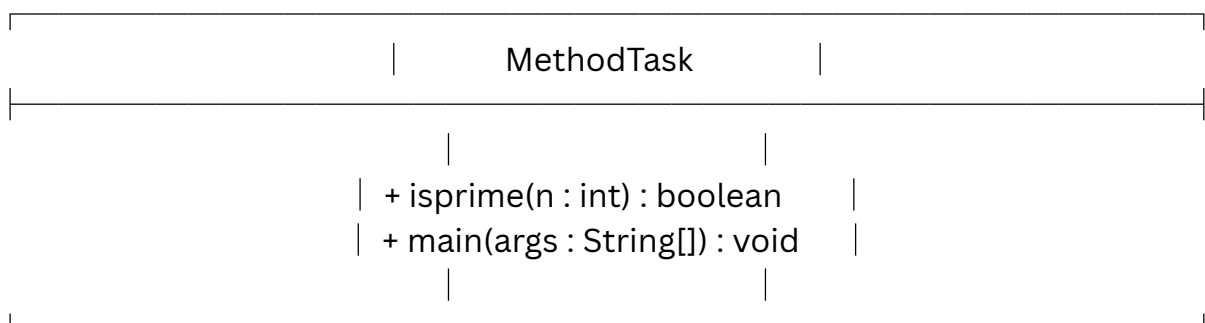
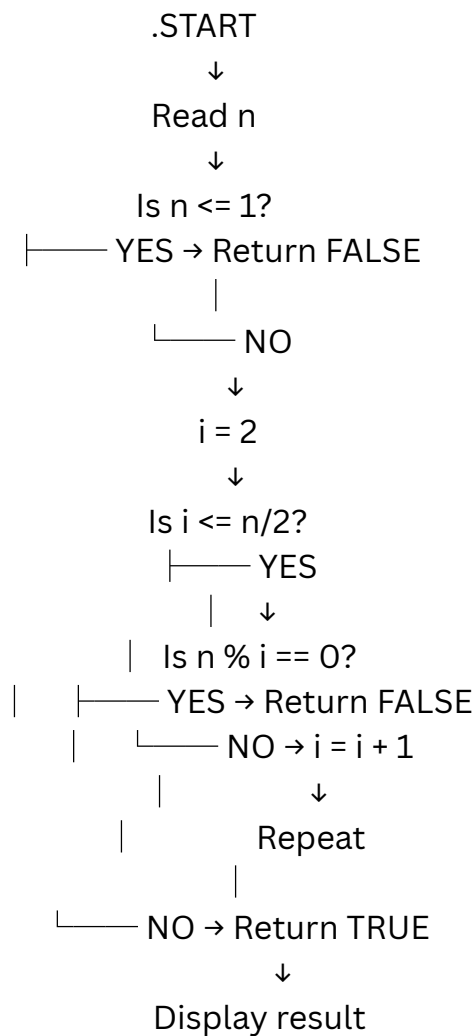
enter the number

5

factorial = 120

Steps

Start the program.
Import Scanner.
Create method isprime(int n).
If $n \leq 1$, return false.
Check divisibility from 2 to $n/2$.
If n is divisible by any number, return false.
Otherwise return true.
Read the number in main().
Call isprime(n).
Display whether the number is prime or not.
Stop



↓
STOP

```

import java.util.Scanner;

public class MethodTask {

    static boolean isprime(int n){
        if(n <= 1)
            return false;
        for (int i =2 ; i<= n/2 ;i++){
            if(n %i == 0)
                return false;
        }
        return true;
    }

    public static void main(String[] args) {

        Scanner pr = new Scanner(System.in);
        int n;
        System.out.println("enter the nnumber");
        n = pr.nextInt();
        if(isprime(n))
            System.out.println("PRIME NUMBER");
        else
            System.out.println("not a prime number");
    }
}

```

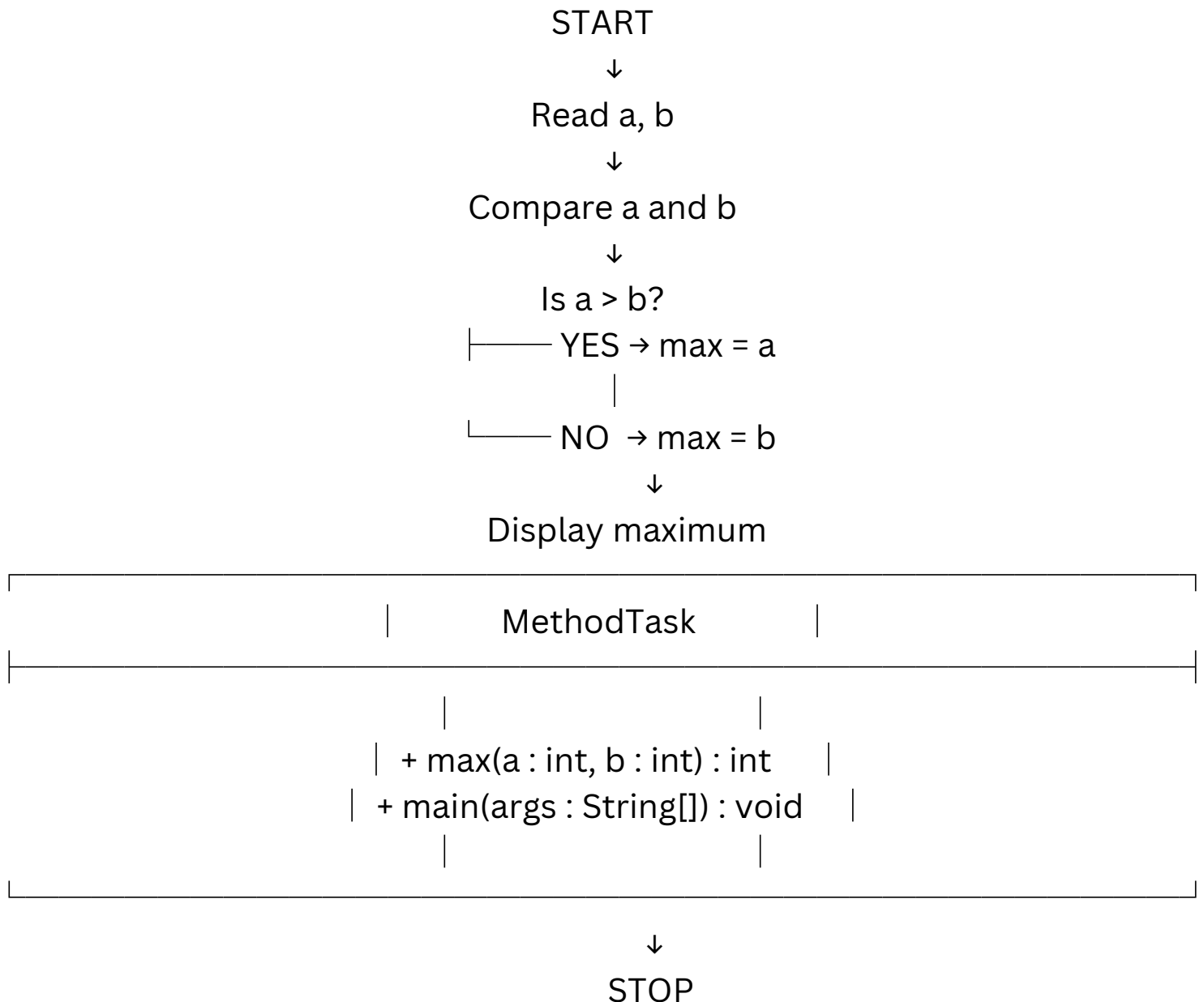
enter the nnumber

7

PRIME NUMBER

Maximum of Two Numbers using Method Steps

Start the program.
Import Scanner.
Create method max(int a, int b).
Compare a and b.
If $a > b$, return a.
Otherwise return b.
Read a and b in main().
Call max(a,b).
Store the returned value.
Display the maximum value.
Stop



```
import java.util.Scanner;
public class MethodTask {

    static int max(int a , int b){
        if(a>b)
            return a;
        else
            return b;
        }

    public static void main(String[] args) {
        Scanner mt = new Scanner(System.in);
        int a ,b , max;
        System.out.println("enter the values of a,b");
        a = mt.nextInt();
        b = mt.nextInt();
        max = max(a,b);
        System.out.println("maximum is " + max);
    }
}
```

enter the values of a,b

25

40

maximum is 40

Area of Circle using Method

Steps

Start the program.

Import Scanner.

Create method area(double radius).

Calculate area using the formula:

$$\text{Area} = \pi \times r \times r$$

Return the calculated area.

Read the radius from the user.

Call area(radius).

Store the returned value.

Display the area.

Stop.

START



Read radius



Calculate

$$\text{Area} = 3.14 \times \text{radius} \times \text{radius}$$



Return Area



Display Area

MethodTask

+ area(radius : double) : double

+ main(args : String[]) : void

STOP

```
import java.util.Scanner;

public class MethodTask {
    static double area(double radius){
        return 3.14 * radius *radius;
    }
    public static void main(String[] args) {
        Scanner r= new Scanner(System.in);
        double radius , result;
        System.out.println("enter the radius");
        radius = r.nextDouble();
        result = area(radius);
        System.out.println("area = " + result);

    }

}
```

enter the radius

5

area = 78.5